

World Development Indicators Analysis (2022)

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This report analyses a sample of the World Development Indicators (WDI) dataset for 2022. The data source is the World Bank WDI database (World Bank 2022).

Exploratory Data Analysis

We focus on three key indicators for 2022: GDP per capita, life expectancy, and inflation rate. These variables capture economic development, health outcomes, and macroeconomic stability.

```
df[["gdp_per_capita", "life_expectancy", "inflation_rate"]].describe()
```

	gdp_per_capita	life_expectancy	inflation_rate
count	203.000000	209.000000	169.000000
mean	20345.707649	72.416519	12.493936
std	31308.942225	7.713322	19.682433
min	259.025031	52.997000	-6.687321
25%	2570.563284	66.782000	5.518129
50%	7587.588173	73.514634	7.967574
75%	25982.630050	78.475000	11.665567
max	240862.182448	85.377000	171.205491

```
df[["gdp_per_capita", "life_expectancy", "inflation_rate"]].corr()
```

	gdp_per_capita	life_expectancy	inflation_rate
gdp_per_capita	1.000000	0.638609	-0.169742
life_expectancy	0.638609	1.000000	-0.111709
inflation_rate	-0.169742	-0.111709	1.000000

The summary statistics show that GDP per capita is highly dispersed, with large variation across countries. Life expectancy varies within a narrower range, while inflation exhibits strong right-skewness due to several extreme outliers.

The correlation matrix indicates a positive relationship between GDP per capita and life expectancy, suggesting that higher-income countries tend to experience better health outcomes. Inflation appears weakly related to the other two indicators.

Visualizing Key Relationships

The following plots illustrate the patterns identified in the exploratory analysis.

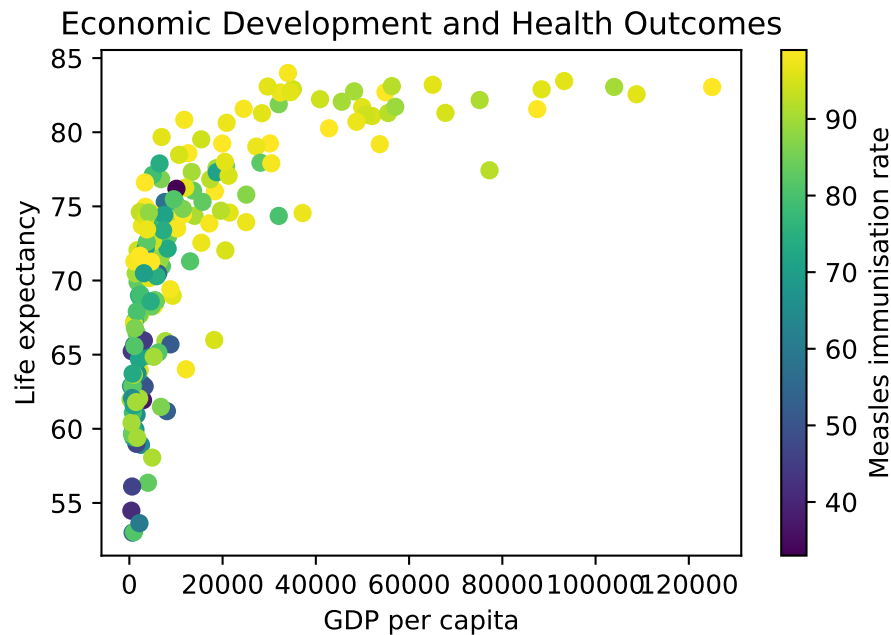


Figure 1: GDP per capita vs Life Expectancy, colored by measles immunisation rate (2022).

Figure Figure 1 visually confirms the positive association between income and life expectancy identified in the correlation analysis.

Figure Figure 2 illustrates the right-skewed distribution of inflation, consistent with the presence of extreme outliers observed in the summary statistics.

Summary of Key Indicators

Indicator	Main Observation
GDP per capita	Highly dispersed across countries
Life expectancy	Positively associated with GDP
Inflation rate	Right-skewed with extreme outliers

Indicator	Main Observation
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Table **?@tbl-key-stats** summarises the main statistical patterns identified in the exploratory analysis.

The positive relationship between income and life expectancy is consistent with classic findings in the development literature (Preston 1975).

References

- Preston, Samuel H. 1975. “The Changing Relation Between Mortality and Level of Economic Development.” *Population Studies* 29 (2): 231–48.
- World Bank. 2022. “World Development Indicators.” <https://data.worldbank.org/indicator>.

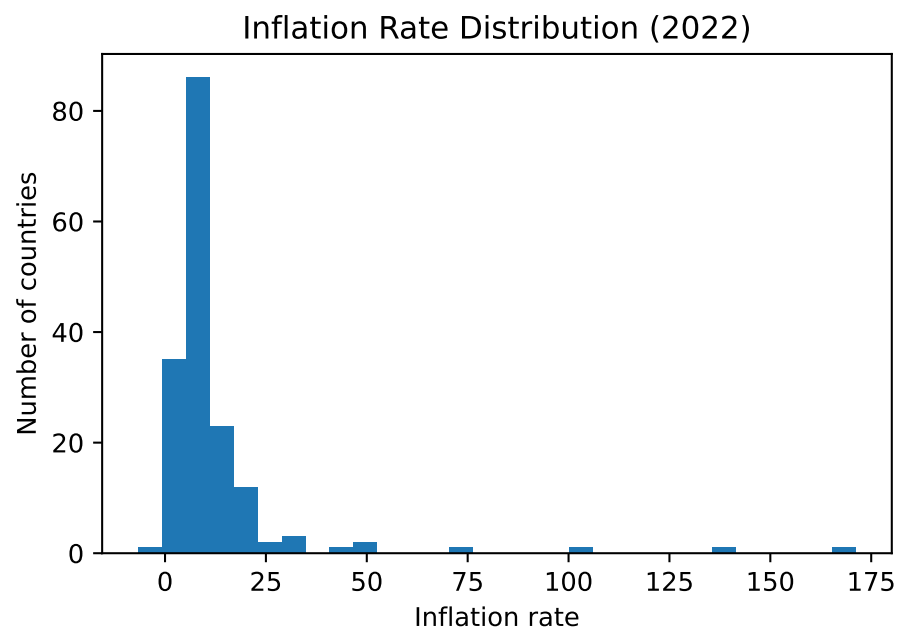


Figure 2: Distribution of inflation rates across countries (2022).